

About applying AODV and OLSR routing protocols to relaying network scenario in FANET with mini-UAVs

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Abstract– Ad Hoc Network is one of the promising technologies for constructing communication networks. Self-organization principle underlies such wireless networks as WSN (Wireless Sensor Networks), MANET (Mobile Ad Hoc Networks), VANET (Vehicular Ad Hoc Networks), and FANET (Flying Ad Hoc Networks). FANET is concerned with the deployment of multiple mini-UAVs. FANETs possess some unique characteristics, if compared to other kinds of ad hoc networks, and develop some major challenges to the research community. One of the problems is designing FANET routing protocol. The protocol should form an effective route between UAVs and adapt it to rapidly changing topology in real time. The performance of AODV and OLSR routing protocols in mini-UAV ad hoc networks is investigated with ns-2 simulation tool. AODV and OLSR routing protocols were studied in the article to choose the best for flying ad hoc network of mini-UAVs. This article describes real possible FANET application of relaying network scenario and describes available mobility model for simulators. For the research purpose the network structure imitation model was built in ns-2.

Index Terms –UAV, FANET, AODV, OLSR, ns-2.

I. INTRODUCTION

ADHOC Network is one of the promising technologies for constructing communication networks. Just like other networks, it is designed to provide access (network connection) and traffic transit functions. In fact, Ad Hoc Network is a decentralized self-organizing network. Its nodes support not only the link with their neighbors, but they also perform the function of forwarding the passing traffic. Self-organization principle underlies such wireless networks as WSN (Wireless Sensor Networks), MANET (Mobile Ad Hoc Networks), VANET (Vehicular Ad Hoc Networks), and FANET (Flying Ad Hoc Networks) [1]. For the past few years the role of unmanned aerial vehicles (UAVs) has been increasing in various areas, such as providing temporary link for remote and inaccessible area in an emergency situation, air monitoring of the ground area in survey and search-and-rescue missions with transmission of video-data and images to the ground station in real time [2]. Consequently, it can be inferred that in a multi-sender mobile network of UAVs, according to the mission requirements, another research problem may emerge, such

as multi-hop connectivity and routing in aerial networks. These communication requirements are essential to establish a reliable aerial network.

II. THEORY

In this regard, since recently the world scientific community has been investigating the concept of utilizing UAVs group, forming Flying Ad Hoc Network (FANET), capable of coordinated completion of one or more assigned tasks. Such group consists of relatively inexpensive mini-UAVs of multicopter type with an electric motor unit (Fig. 1) [3].

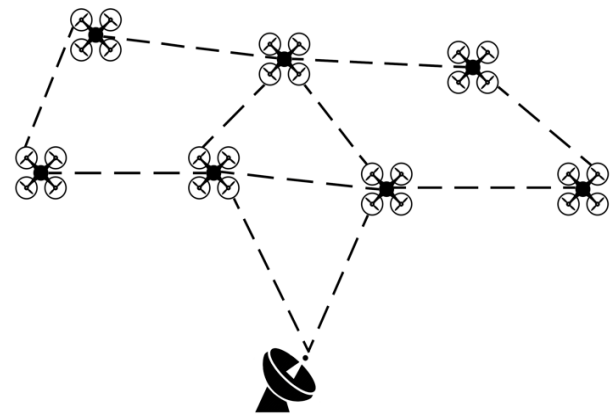


Fig. 1. Flying Ad Hoc Network (FANET).

A. FANET Characteristics

As in MANETs, the FANET architecture is a network without infrastructure, using multiple nodes to transmit data packets. The other characteristics are self-organized abilities, self-managed data in a distributed manner, the nodes communications and cooperation that are used to deliver data. FANETs, however, have some specific features as well that makes them differ from MANET and VANET [4]:

1) Network connectivity: FANET network connectivity has higher intermittent degree than in MANETs or VANETs. It is caused by UAVs mobility. The communication interruption can be critical when important information (control/command traffic) is transmitted. Moreover, UAV failure may result in connectivity failure

with a following routing failure, and subsequent communication failure or worse delay.

2) Nodes number: When a UAV with a relatively high velocity is deployed in a specific mission, a restricted mission area can be covered. In this case, a large number of UAVs is not necessary.

3) Sufficient energy: It is usually supposed that depending on their size and type, FANET nodes have sufficient energy and computing power if compared to MANET nodes. The reason is that the amount of energy needed to move an UAV is much greater than that required to compute data.

4) Mobility (3D): The mobility model is of importance when network protocols are designed for ad-hoc networks. The mobility patterns of UAVs and other vehicles differ. UAV can move in all three dimensions.

5) Environment: Since FANETs consist of flying nodes, they are usually used in large free spaces. Therefore, to model the physical layer one often uses the free-space path loss model. Still large obstacles, ground reflections or weather conditions and other factors can affect UAVs connectivity. These factors should be taken into consideration in the applied propagation model.

III. PROBLEM DEFINITION

A. On The Routing In FANET

To perform FANET missions, control and data traffic must be relayed by UAVs and GCS. The routing protocols used in FANET should meet the requirements considering the automatic search for the best route (route group) to guarantee one or more qualitative parameters of the data transmission/reception process. Therefore, the search for the route (or route group) among the network nodes is one of the major tasks in FANET. Moreover, the route parameters, such as connectivity (the route quality), route length (the number of hops/iterations in the route) and a share of transit nodes (a share of the network nodes involved in routing), affect the network traffic performance [5].

During the investigation, the most widespread and widely used routing protocols in FANET were used on the network level, that is reactive protocol AODV (Ad hoc On-Demand Distance Vector) and proactive protocol OLSR (Optimized Link-State Routing) [6].

B. Proactive Routing Protocol, OLSR

Proactive routing means that each node provides fresh information about the network state to all the nodes before transmitting data packets. Therefore, each node is required to relay control packets systematically. Consequently, every node always has a route to connect to any other node in the network. As a rule, a routing table is necessary for the nodes to be able to create routes due to the predefined criteria.

OLSR protocol maintains a routing table at each network node by gathering topology information through Topology Control (TC) messages. HELLO messages are also used to find one hop and two hop neighbors. An important feature for reducing control messages in OLSR is multi point relaying. Multi Point Relay (MPR) nodes are a subset of nodes responsible for forwarding link state updates. This optimization to pure link state routing protocol proves beneficial in highly dense environments where MPR mechanism is efficiently utilized [7].

The standard OLSR parameters are specified without definite values for their ranges. The standard OLSR parameter values are given in Table I, as stated in the OLSR RFC 3626.

The range of possible values for each parameter has been determined by following OLSR restrictions to avoid pointless configurations [8] [9].

TABLE I
MAIN OLSR PARAMETERS AND RFC 3626 STATED VALUES

Parameter	Standard Configuration	Range
HELLO_INTERVAL	2.0 s	$\mathbb{R} \in [1.0, 30.0]$
REFRESH_INTERVAL	2.0 s	$\mathbb{R} \in [1.0, 30.0]$
TC_INTERVAL	5.0 s	$\mathbb{R} \in [1.0, 30.0]$
WILLINGNESS	3	$\mathbb{Z} \in [0, 7]$
NEIGHB_HOLD_TIME	3*HELLO_INTERVAL	$\mathbb{R} \in [3.0, 100.0]$
TOP_HOLD_TIME	3*TC_INTERVAL	$\mathbb{R} \in [3.0, 100.0]$
MID_HOLD_TIME	3*TC_INTERVAL	$\mathbb{R} \in [3.0, 100.0]$
DUP_HOLD_TIME	30.0 s	$\mathbb{R} \in [3.0, 100.0]$

C. Reactive Routing Protocol, AODV

In this type of routing protocol, a route is formed only when a node needs to transmit a packet to a destination node. At first a node sends a route request packet to search for a route. This packet is broadcasted through the network. When the route request packet is received, a node puts its address to the list of nodes that have already processed the packet. When one (or several) route request packets arrive at the destination node, it sends back a route response packet along the shortest discovered route. This route is used by the source node to reach the destination.

AODV routing protocol is a reactive protocol which determines route to destination only when source initiates it and keeps it as long as the source desires. A route request (RREQ) packet is broadcasted by the source node to discover route to the destination. The intermediate nodes not only forward RREQ but also update themselves with the source information as contained in RREQ thus setting up a reverse route entry to the source. Any intermediate node that has route to the destination replies the source with route reply (RREP) packet containing the number of hops required to reach the destination. In case of an invalid route, a route error packet (RERR) is generated to inform the source about link failure so that it can re-start the route discovery process. In terms of overhead AODV dominates its proactive counterparts as routes are discovered only on demand. However, it suffers from latency issues since a

packet has to wait till the route to the new destination is found [10].

A set of the major AODV parameters with their default values as given in RFC 3561 are presented in Table V. The range of possible values for each parameter has been established here by following AODV restrictions in order to avoid pointless configurations [11].

TABLE II
MAJOR AODV PARAMETERS. DEFAULT VALUES FOLLOWING
THE RFC3561 SPECIFICATION

Parameter	Default Values	Range
ACTIVE_ROUTE_TIMEOUT	3.0 s	1.0 ... 10.0
ALLOWED_HELLO_LOSS	2 HELLO packets	1 ... 10
MY_ROUTE_TIMEOUT	2.0 *ACTIVE_ROUTE_TIMEOUT	1.0 ... 10.0
NET_DIAMETER	35 nodes	1 ... 50
NODE_TRAVERSAL_TIME	0.04 s	0.01 ... 1.0
NET_TRAVERSAL_TIME	2.0 *NODE_TRAVERSAL_TIME *NET_DIAMETER	1.0 ... 10.0
RREQ_RETRIES	2 tries	1 ... 10
RREQ_RATELIMIT	10.0 kbps	1.0 ... 10.0
TTL_START	1.0 s	1.0 ... 10.0
TTL_INCREMENT	2.0 s	1.0 ... 10.0
TTL_THRESHOLD	7.0 s	1.0 ... 20.0

IV. PROPOSED METHODOLOGY

A. Relaying Network

FANETs can deploy connectivity in areas that need a certain kind of communication. Autonomously operated UAVs are being used as airborne communication relays to efficiently and securely transmit information and for increasing the communication range of relaying ground nodes (Fig 2) [12].

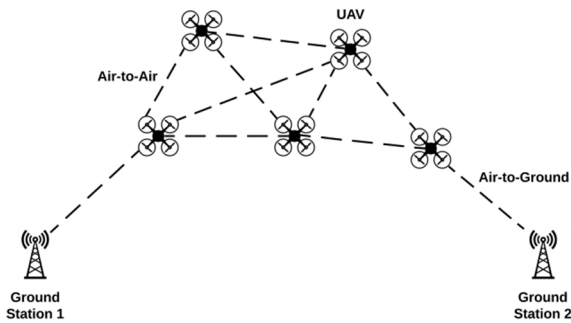


Fig. 2. Example use of FANET in relaying network application

Typically, the mobility of such backbones is very low, since there is no particular reason why UAVs should move in space, as well as to ensuring connectivity. In fact, the more static are the mini-UAV, the better is the network

performance. There are cases where the end nodes of the backbones must follow moving targets to guarantee communication with them (e.g. rescue personnel or victims in disaster areas), but the mobility is still limited to the end nodes and is characterized by low speeds. If the targets to be covered are moving fast, like in the case of maintaining connectivity among vehicles.

B. Network Traffic

The definition of traffic flows in the FANET should also be such that it can be as close as possible to a realistic mission. In our simulations, two nodes of the scenario are called ground station control (GSC). These nodes communicate with each other bi-directionally. Every other node in the network has uni-directional flows towards each of these nodes.

Considering such traffic flows are compatible with a FANET scenario in which several GSCs are receiving data from UAVs nodes. Considering such traffic flows are compatible with a FANET scenario in which several GSCs are receiving data from UAVs nodes. In order to have a realistic comparison of the protocols, assuming such an environment is required. The other important fact to use such flows in the network is that we can test adaptability of the routing protocols.

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C. Propagation Model

The reason for selecting this model over shadowing model is that in FANETs there is a clear LoS (Line of Sight) between the transmitter and receiver UAVs [6]. It is assumed that radio signal transmission-reception is stable and all UAVs transceivers are set up identically. The network nodes are without obstacles, and without due consideration of multiple possible signal transmission paths and refractions. To determine transmission range in open environment and free space, Friis equation is used [13]:

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi d)^2 L} \quad (1)$$

where:

d is the distance between transmitting and receiving antennae;

P_t is the transmitting antenna power at the distance d ;

P_r is the power received by the antenna;

G_t is the gain ratio of the transmitting antenna;

G_r is the gain ratio of the receiving antenna;

λ is the wavelength.

L is the system loss, $L = 1$ in ns-2 simulations.

D. MAC/PHY Specifications

The propagation model considered in our simulations is a free space path loss, which models the propagation as a disc around the transmitter. Although channel modeling of FANETs can be a research topic based on the specific channel characteristics imposed by UAV mobility and the environment, the effects of channel impairments are not addressed in this research. Instead, the focus of this article is on routing protocol design, thus assuming a simple channel model with a predictable set of specifications is acceptable.

To simulate the Atheros AR9344 [14] radio with direct-sequence spread spectrum, the physical radio characteristics of the network interface for each mobile node were selected, including the receiver sensitivity, the transmission power, and the antenna gain. Based on the packet reception power threshold, which is -96 dBm, the transmission range of the UAVs will be 500 m. Transmit power for acquiring such a range is 15dBm. The values for such a setting are assumed to be typical values for mini-UAVs.

E. Mobility Model

Group mobility models can be used to simulate group of UAVs in performing autonomous operations without centralized control. Based on the previous study by the authors who investigated the performance of different mobility models for FANETs, Random Way Point (RWP) has been used in this article [15].

An example of a mobility pattern using RWP is shown in Fig. 3.

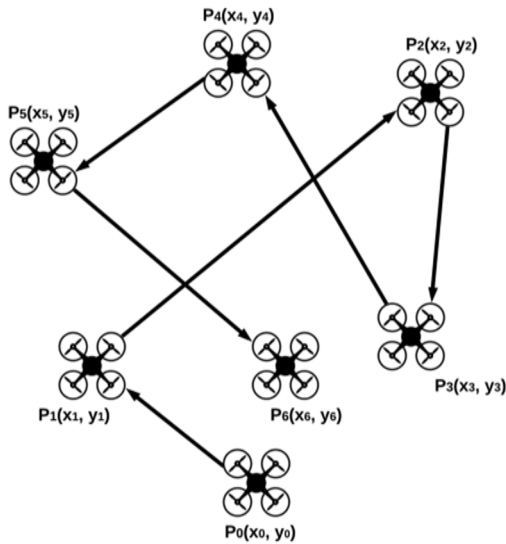


Fig. 3. Random waypoint (RWP) mobility model.

An important part of the simulation configuration is to have suitable mobility scenarios, which represent plausible FANET deployments. In general, proposing accurate mobility scenarios for FANETs is an open research problem (which is beyond the scope of this research). In

this article, the attempt was to use one of the available NS-2 mobility models, which can be adapted to fit searching and tracking missions by changing several NS-2 settings. The mobility scenarios are derived from the RWP model.

The assumption in a search mission is that every UAV is looking at different places to determine the desired object. Therefore, the RWP can model the mobility of the FANET, especially when UAVs move independently (in fact, our model does not consider UAVs' statistical interdependencies). In such a case, a node chooses a destination and speed, and then moves from its current location at that speed towards the destination. We consider a continuous flight mission in which the UAVs never come to a rest, which is why pause time is set to 0. Also, the mission starts at time 0 and ends at $t=1800$ sec, which is the end of the simulation.

The shown FANET example includes 50 UAVs moving around the area of 3600×1200 m². A rectangular space was selected to encourage the longer routes usage between the nodes than could occur in a square space with the node density being equal. During the simulation, UAVs exchange data. The velocity of the UAVs is varied from 0 m/s to 60 m/s.

F. Traffic Generation Model

The data flow model does 25 runs of a data generator with constant bitrate (CBR), operating over the user datagram protocol (UDP) agents that are determined in the nodes (UAVs). CBR connections are chosen where source and destination nodes are selected randomly 50 nodes respectively. CBR traffic simulation lasts from 0 to 1800 seconds. The sizes of the CBR data packets are 64, 256, 512 and 1024 bytes (RFC 2544 and RFC 6815), and the packet rate is 4 packets/s [16].

A fixed data rate has been chosen because the aim is not to study the maximum throughput, but to investigate the ability of AODV and OLSR to find and maintain the routes in the given FANET.

TCP sources were not applied as TCP offers the network a conforming load, which means it alters times of sending packets based on its evaluation of the network ability to transmit packets. As a result, the protocols differ both in the times of data packet origination performed by its sender and in the node position when the packet is transmitted, thus making a direct comparison between them impossible [17]. Other details of the simulation parameters are given in Table III.

G. Experimental Setting

The simulation phase is performed by running ns-2 simulator (version ns-2.34) based on the UM-OLSR (ver. 1.0 patch) realization of OLSR. Each optimization method was run 5 times independently on Lenovo ThinkStation P320 SFF with Intel Xeon vPro E3-1245 v6, 16 GB of RAM, and O.S. Ubuntu Desktop 16.04.3 LTS 64-bit.

A different UAV mobility and interactions are used in every independent iteration, based on single random sources inside each simulation, hence contributing to the results generalization.

TABLE III
SIMULATION PARAMETERS IN NS-2

Parameter	Value
Simulation time	1800 s
Simulation area	3600×1200 m ²
Number of UAVs	50
UAV speed	0-60 m/s
Pause time	0 s
Propagation model	Friis
Radio frequency	2.437 GHz
Channel bandwidth	6 Mbps
Antenna model	Omni
Mac protocol	802.11g
Transmission range of UAVs	500
Routing protocol	OLSR, AODV
Packet size	64, 256, 512, 1024 bytes
Transport protocol	UDP
CBR data flow	25

H. Performance Metric

The following eight typical performance measures are given in this section. Moreover, quality of service (QoS) metrics is employed to evaluate the performances of the FANET routing protocols [18].

1) Average Packet Delivery Ratio (PDR) demonstrated a ratio between the number of datagrams received by the destination and that transmitted by the source. This metric is measured on the basis of application layer sequence numbers added in the simulated UDP datagrams:

$$PDR_{ave} = \sum_{i=1}^n \frac{|P\gamma_i|}{|p_i|} \quad (2)$$

where $P\gamma_i$ is the number of datagrams received by the destination, p_i is the number of datagrams transmitted by the source. The bigger PDR, the better the protocol performance.

2) Average End-to-End Delay (EED): It includes all delays caused by buffering during the route discovery, queuing at the interface, retransmission at the Medium Access Control (MAC), propagation and transfer times. To put it simply, the average End-to-End delay is the time necessary for a packet to pass the network from the source to the destination.

$$EED_{ave} = \frac{\sum_{p_i \in P\gamma} T_a(p_i) - T_d(p_i)}{|P\gamma_i|} \quad (3)$$

where p_i is the current packet, $T_a(p_i)$ is the reception time for p_i and $T_d(p_i)$ is the time of the packet sending p_i . The smaller EED, the better the performance of the protocol.

3) Normalized routing overhead (NO): The ratio of total number of routing packets generated to that of data packets received during the simulation time.

$$NO = \sum_{i=1}^n \frac{|M_i|}{|P_i|} \quad (4)$$

where $M_i \subset P$ and $|M_i|$ are the total numbers of control packets required for route discovery. The lower, the better.

4) Routing Overheads (RO) were calculated as all control messages of each routing protocol in bytes divided by the simulation time:

$$RO = \frac{\sum_{i=1}^{i=n} |M_i|}{T} \quad (5)$$

where $M_i \subset P$ and $|M_i|$ are protocol control message sizes in kbit/s; T is the control messages number in the simulation. The lower, the better.

5) Average Hop Count (HOP) was determined on the basis of Time-to-Live field in the IP-header (for AODV and OLSR) in the datagrams received by the destination, then summarized and divided by the received datagrams number:

$$HOP_{ave} = \frac{\sum_{p_i \in P\gamma} H(p_i)}{|P\gamma_i|} \quad (6)$$

where $H(p_i)$ is all hops performed by datagrams p_i before the destination is reached. The smaller, the better.

6) Average Throughput is all UDP payload arriving at the destination during the simulation and divided by the simulation time:

$$Throughput_{ave} = \frac{\sum_{i=1}^{i=n} |P\gamma_i|}{T} \quad (7)$$

where $P\gamma_i$ is all successfully received datagram payload size in kbit/s; T is the given time period where the data packets successfully arrive at the same destination. The bigger throughput, the better the performance of the routing protocol.

7) Jitter: Jitter is the variation of the delay between every received data packets. The packet arrival time should vary minimally in order to have better performance in Mobile Ad-hoc Networks (RFC5148). Jitter value is calculated according to RFC 3550. Jitter J_i for i th packet is determined by the formula:

$$J_i = J_{i-1} + \frac{|D_{i-1}| - J_{i-1}}{16} \quad (8)$$

where D_i is deviation from the expected arrival time for the i th packet.

Deviation from the expected arrival time for the i th packet D_i is calculated by the formula:

$$D_i = (R_i - R_{i-1}) - (S_i - S_{i-1}) \quad (9)$$

where R is the arrival time in UDP time tags, while S is UDP time tag taken from the packet.

8) Dropped Packets (DP): In a computer network, packet loss occurs when one or more packets moving across a computer network do not reach their destination. The total number of packets dropped during the transmission is calculated as follows.

$$DP = \sum_{i=1}^n P_i - \sum_{i=1}^n P\gamma_i \quad (10)$$

where P_i is all packets generated by the senders, $P\gamma_i$ is all packets successfully arriving to the receiver.

V. FINDINGS AND DISCUSSION

AODV and OLSR routing protocols experimental analysis have been performed during the simulation and the results obtained are shown in tables and on their results graphs are drawn.

A. Average Packet Delivery Ratio

AODV packet delivery ratio is observed to be consistent and almost near to hundred percent whereas OLSR packet delivery ratio decreases when the node speed increase (Fig.4 and Fig.5).

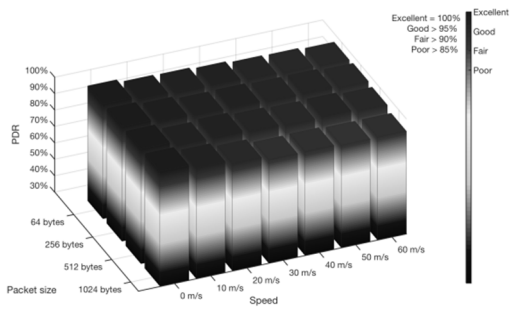


Fig. 4 Packet delivery ratio vs Speed vs Packet size for AODV

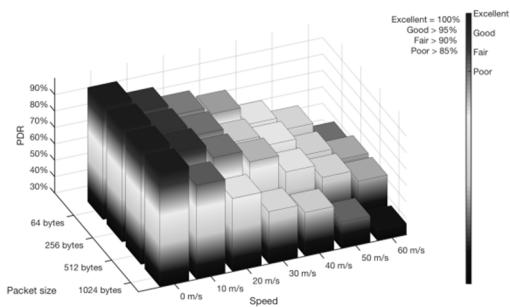


Fig. 5 Packet delivery ratio vs Speed vs Packet size for OLSR

Both reactive and proactive routing protocols have similar nature with increase in speed. However, the PDR decreases significantly at higher speeds for OLSR. Due to frequent topology change at high speeds, the number of stale routes increases in the routing tables of the proactive protocols and the broken link rediscovery takes time. The dips observed at certain speeds are due to the random

scenario generation used, however, the average results show decreasing trend.

B. Average End-to-End delay and jitter

At low node speed end to end delay of AODV is more than OLSR but as the speed of nodes increases the end to end delay time of AODV routing increases (Fig. 6). Whereas OLSR end to end delay time increases a little. So in case of end to end delay OLSR perform better than AODV routing protocol.

OLSR shows the lowest average End-to-End delay and jitter in the network. Being table driven protocol, these protocols do not encounter long route setup time as updated routes are presented in the routing tables. On the other hand AODV does not reuse routing information and has to introduce the route discovery process again and again when a node is to be transmitted.

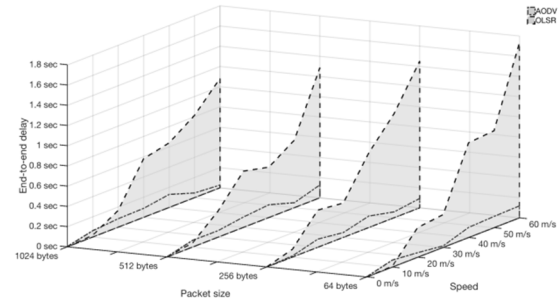


Fig. 6. Average End-to-End delay vs Speed vs Packet size

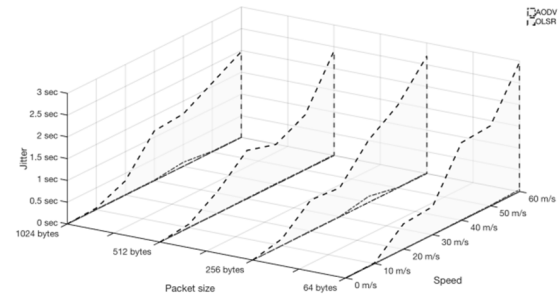


Fig. 7. Jitter vs Speed vs Packet size

By the time the route reply message reaches the source in response to route request query, the destination has moved away due to high mobility. Hence, in the time-sensitive network topology AODV shows poor delay bounds as compared to proactive protocol OLSR.

C. Average Throughput

The average throughput of AODV is consistent even when the node speed increases, but average throughput of OLSR is getting down with the high speed nodes (Fig. 8).

D. Routing Overheads

Routing Overhead and Normalized Routing overheads for AODV and OLSR are in the network (Fig. 9 and Fig.

10). OLSR shows high routing load due to massive generation of Topology Control (TC) and HELLO packets to maintain the routing table. OLSR computes Multi-Point Relays (MPRs) through the TC and HELLO messages. MPRs are selected to access two hop neighbors within the network and reduce broadcast overhead but due to less node density and random movement of nodes within the cluster in the network, MPR mechanism is inefficiently used. Although AODV does not generate periodic topology update messages but the dynamic topological behaviour of the network causes AODV to broadcast frequent route request messages which results in high routing overhead.

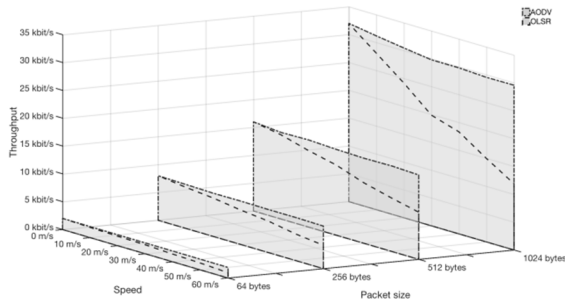


Fig. 8. Average Throughput vs Speed vs Packet size

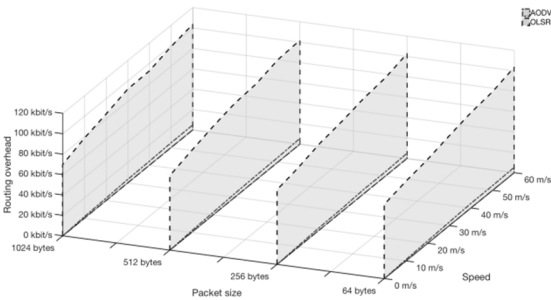


Fig. 9. Routing overhead vs Speed vs Packet size

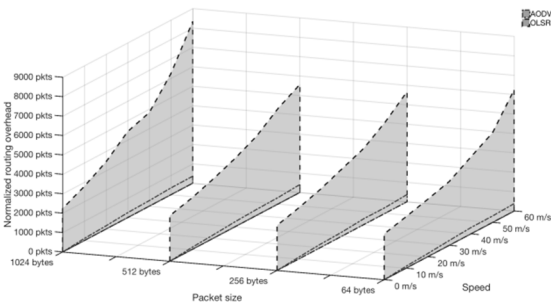


Fig. 10. Normalized routing overheads vs Speed vs Packet size

E. Average Hop count

Hop count metric for AODV and OLSR for our network are shown in Fig. 11.

The average hop count of OLSR and AODV are about 5-5.7 hops and 6.9-6.3 hops, respectively.

F. DropPackets

There are several reasons of packet loss in FANET. Dropped packets are significant issues and many factors may be reasons, e.g. network topology, mobility, link interference and network congestion.

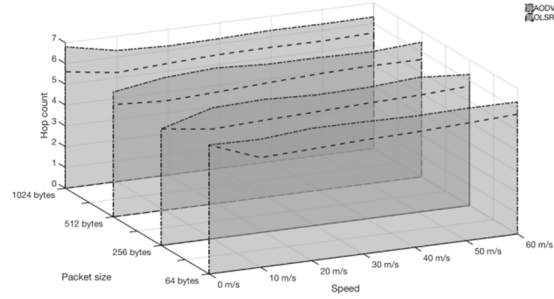


Fig. 11. Average hop count vs Speed vs Packet size

Several reasons of packet drops in NS2 wireless model have been examined. Some reasons of drop actions in NS2 are explained in Table IV [19].

TABLE IV
EXPLANATION OF PACKET DROPS ON NS-2

	Explanation of Packet Drop
MAC LAYER	
COL	Collusion
DUP	Duplicate
ERR	MAC Error
STA	MAC State invalid
BSY	MAC Busy
RET	Retry Count Exceed
ROUTER LAYER	
LOOP	Loop on Route
TOUT	Time Out
IFQ	Interface Queue Full
ARP	ARP Queue Full
OUT	Outside Subnet
CBK	Call Back
TTL	Time to Live
NRTE	No Route Exceed
GENERAL	
END	Simulation End

Fig. 12, 13 prove absence of packet drop due to the ERR, STA, BSY, TOUT and OUT, but still there are very few drops due to the DUP, LOOP, END and COL.

Simulation results show that AODV outperforms OLSR with respect to packet delivery, throughput and routing overheads. Also findings prove that OLSR perform better than AODV with regard to jitter, end to end delay and hop count.

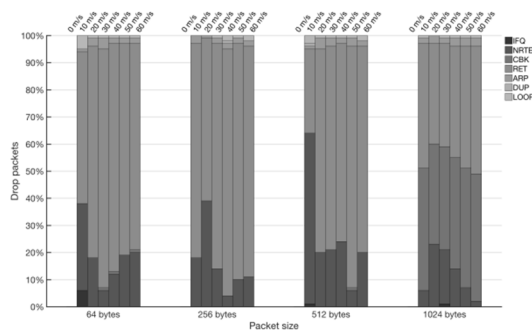


Fig. 12. Drop packets vs Speed vs Packet size of AODV protocol

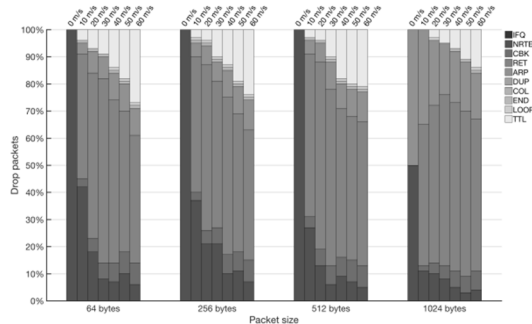


Fig. 13. Drop packets vs Speed vs Packet size of OLSR protocol

VI. CONCLUSIONS

This article identifies performance of AODV and OLSR routing protocols in terms of Packet Delivery Ratio, End to End Delay, Average throughput, routing overheads and etc. The evaluation based on these protocols determines some important behavior and characteristic of these protocols when they are used in FANET.

OLSR routing algorithm is not well adjusted for highly dynamic and low density networks like FANETs due to the large amount of packet overhead generated. The overhead tends to increase proportionally with the number of nodes and the mobility degree. There is also a negative impact on energy level as nodes do not have an idle phase and continuously follow the wireless medium. However, in spite of these drawbacks, proactive routing has an advantage to offer fast connections between nodes as routing information is immediately available in the routing table, so there is no delay when data traffic is to be sent.

AODV outperformed OLSR in terms of throughput, packet delivery and routing overheads in an air-to-air FANET scenario. It is proved with the reactive nature of its topology changes and its limited overhead. Under the scenario, AODV has better performance than OLSR in terms of packet delivery ratio, but it can develop higher delays with low workload. However, this evaluation is based on simulations and does not consider real system related issues.

To develop an efficient routing algorithm for FANETs that can deliver payload traffic and control packets, while ensuring a certain QoS and a high level of security services, is a difficult task.

An article identified three common issues, that can be addressed to improve these existing algorithms and to render them suitable for FANETs. The first is lack of real experiments crucial to assess routing protocols based on realistic requirements. The second issue is lack of security consideration and it is highly significantly due to inherent characteristics of the wireless medium. Lastly, the main drawback is lack of consideration for the routing algorithm certification. FANET certification is mandatory for UAVs to be integrated into civil airspace.

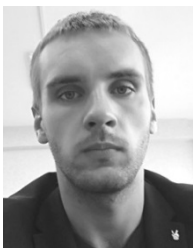
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